

# Day 4 — Building a Tithi Calendar

**Module:** Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

## Learning objective

By the end of this lesson, students will: construct a tithi calendar for the current month using observation and computation.

## Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

## Vocabulary introduced today

- *kṛṣṇa pakṣa* (IAST: kṛṣṇa pakṣa; lit. “waning fortnight (full → new)”)

## Lesson flow

### Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

### Core (20 min)

1. **Pull up a current pañcāṅga** (online or printed). Display the next 14 days with tithi, pakṣa, day-of-week.
2. **Manual construction.** Without looking at the pañcāṅga, students attempt to predict the tithi 7 days from today using:
  - Today’s tithi (e.g. *śukla pañcamī*).
  - Approximate rule: ~1 tithi per 24h.
  - Note: actual tithi may be off by  $\pm 1$  due to varying lunar speed.
3. **Source.** Modern Drik pañcāṅga uses Sūrya-siddhānta-derived corrections plus modern astronomical constants (NASA JPL ephemerides). Show the students that traditional and modern computation agree to within minutes.
4. **The intercalation challenge.** Adhika māsa (added month) occurs roughly every 32 months when no Sun-samkrānti happens within a lunar month. The 2026–2030 adhika māsa calendar is published; pull it up. “*Why is this needed? What would happen without it?*”

5. **Project artefact.** Each student produces a one-month tithi calendar for the CURRENT month, side-by-side with the Gregorian dates. This will be assessed as part of the summative project.

### Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Building a Tithi Calendar.

### Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

### Homework

(See [homework.md](#) for today’s task.)

### Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

### Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).